



PENRITH HIGH SCHOOL

**2016
HSC TRIAL EXAMINATION**

Mathematics Extension 2

General Instructions:

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A Reference Sheet of formula is provided
- In questions 11 – 16, show relevant mathematical reasoning and/or calculations
- Answer all Questions in the writing booklets provided

Total marks–100

SECTION I

Pages 3–7

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

SECTION II

Pages 9–15

90 marks

- Attempt Questions 11–16
- Allow about 2 hours 45 minutes for this section

Student Name: _____

Teacher Name: _____

This paper MUST NOT be removed from the examination room

Assessor: Mr Ferguson

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

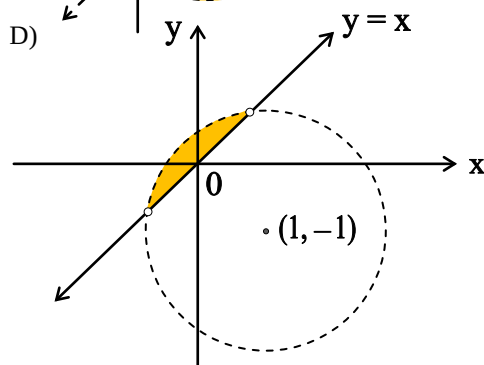
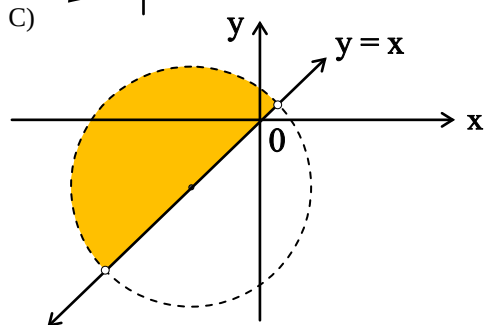
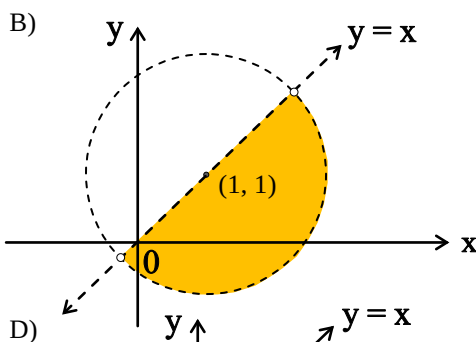
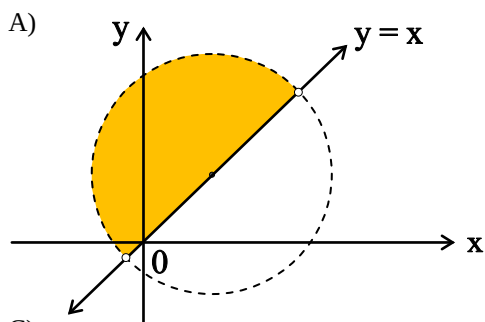
Use the multiple-choice answer sheet for Questions 1–10.

1 What is the modulus and argument of $-1+i$?

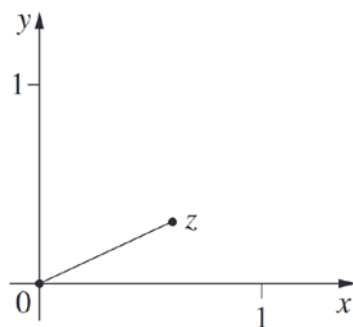
- (A) Modulus $\sqrt{2}$ and argument $\frac{\pi}{4}$
- (B) Modulus $\sqrt{2}$ and argument $\frac{3\pi}{4}$
- (C) Modulus 2 and argument $\frac{\pi}{4}$
- (D) Modulus 2 and argument $\frac{3\pi}{4}$

2 The complex number z satisfies the inequations $|z-1+i| < 2$ and $\text{Im}(z) \geq \text{Re}(z)$.

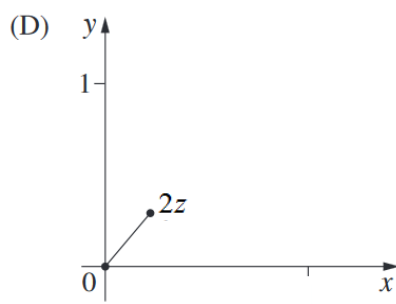
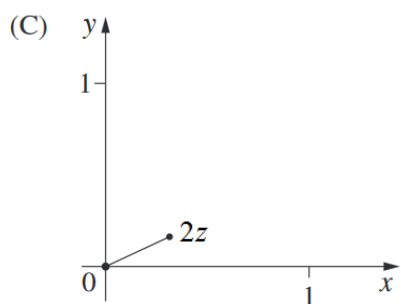
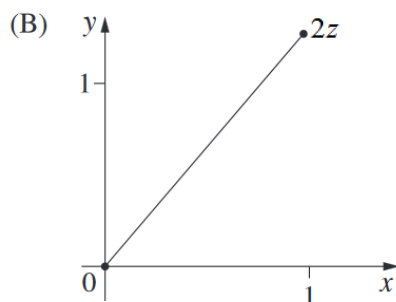
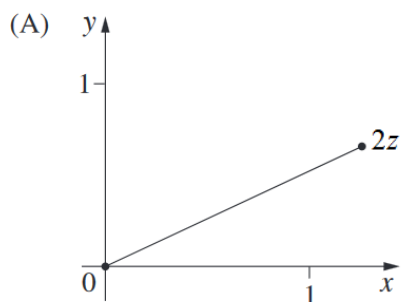
Which of the following shows the shaded region in the Argand diagram that satisfies these inequations?



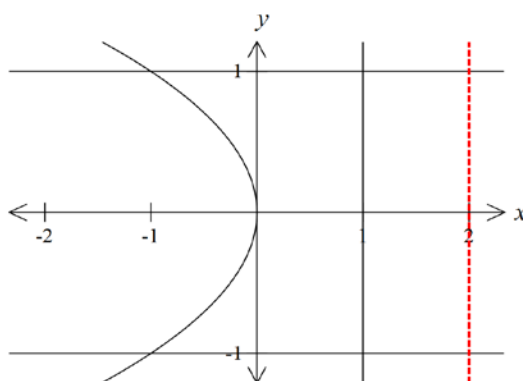
- 3 The Argand diagram below shows a complex number z .



Which diagram best represents $2z$?

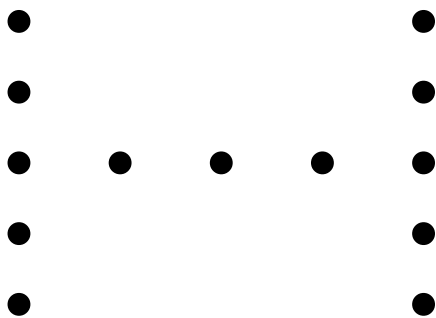


- 4 The region is bounded by the lines $x=1$, $y=1$, $y=-1$ and by the curve $x=-y^2$. The region is rotated through 360° about the line $x=2$ to form a solid. What is the correct expression for volume of this solid?



- (A) $V = \int_{-1}^1 \pi(y^4 - 4y^2 + 3)dy$
 (B) $V = \int_{-1}^1 \pi(y^4 + 4y^2 + 3)dy$
 (C) $V = \int_{-1}^1 \pi(y^4 - 4y^2 + 4)dy$
 (D) $V = \int_{-1}^1 \pi(y^4 + 4y^2 + 4)dy$

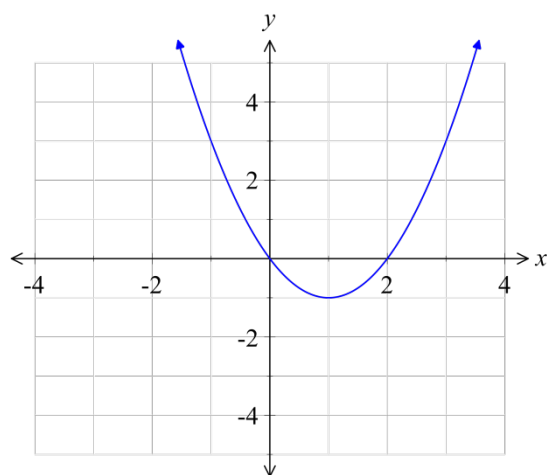
- 5 The foci of the hyperbola $\frac{y^2}{8} - \frac{x^2}{12} = 1$ are
- (A) $(\pm 2\sqrt{5}, 0)$ (B) $(\pm\sqrt{30}, 0)$ (C) $(0, \pm 2\sqrt{5})$ (D) $(0, \pm\sqrt{30})$
- 6 What are the values of real numbers p and q such that $1+i\sqrt{2}$ is a root of the equation $z^3 + pz + q = 0$?
- (A) $p = 1$ and $q = -6$
- (B) $p = -1$ and $q = 6$
- (C) $p = -1$ and $q = -6$
- (D) $p = 1$ and $q = 6$
- 7 The diagram shows a shape made by 13 points.



How many triangles can be made with these points as vertices?

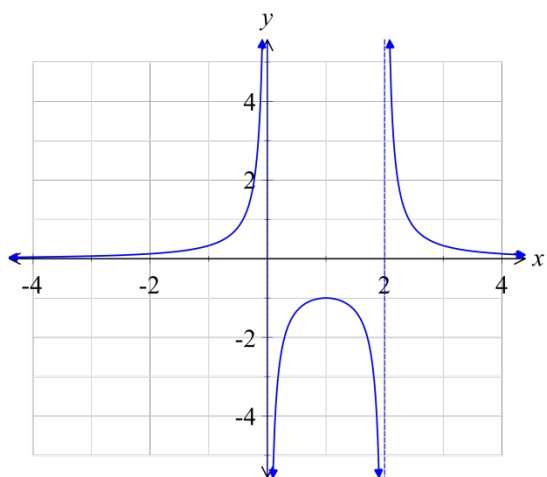
- (A) ${}^{13}C_3 - 3{}^5C_3 - 3$
- (B) ${}^{13}C_3 - 3{}^5C_3 - 4$
- (C) ${}^{13}C_3 - 2{}^5C_3 - 3$
- (D) ${}^{13}C_3 - 2{}^5C_3 - 4$

- 8 The diagram below shows the graph of the function $y = f(x)$.

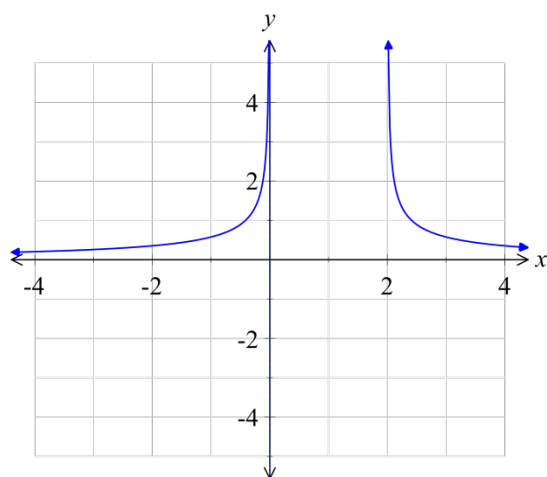


Which of the following is the graph of $y = \frac{1}{f(x)}$?

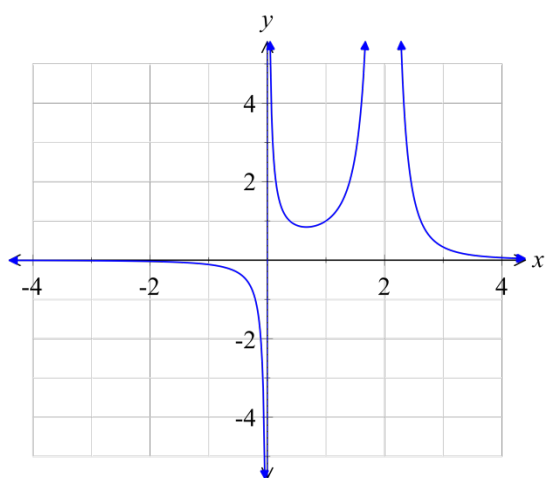
(A)



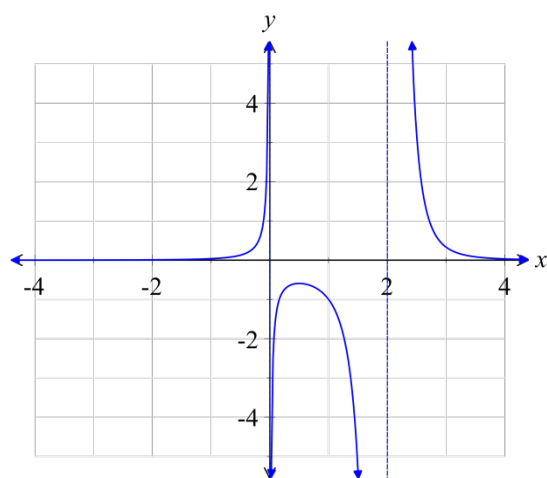
(B)



(C)



(D)



9 Evaluate $\int_0^{\frac{\pi}{4}} x \sec^2 x \, dx$

(A) $\frac{\pi}{4} - \frac{1}{2} \ln 2$

(B) $\frac{\pi}{4} + \frac{1}{2} \ln 2$

(C) $\frac{\pi}{4} + 2 \ln \sqrt{2}$

(D) $\frac{7}{3}$

10 A particle of mass m is moving in a straight line under the action of a force.

$$F = \frac{m}{x^3}(6 - 10x)$$

Which of the following is an expression for its velocity in any position, if the particle starts from rest at $x = 1$?

(A) $v = \pm \frac{1}{x} \sqrt{(-3 + 10x - 7x^2)}$

(B) $v = \pm x \sqrt{(-3 + 10x - 7x^2)}$

(C) $v = \pm \frac{1}{x} \sqrt{2(-3 + 10x - 7x^2)}$

(D) $v = \pm x \sqrt{2(-3 + 10x - 7x^2)}$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question on a new writing sheet. Extra writing sheets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing sheet.

- a) Let $z = 3 - i$ and $w = 2 + i$. Express the following in the form $x + iy$, where x and y are real numbers:

i) $\frac{z}{w}$ 2

ii) $\overline{-2iz}$ 2

b) Let $z = \frac{1}{2} + \frac{\sqrt{3}}{2}i$.

i) Express z in modulus-argument form. 2

ii) Show that $z^6 = 1$ 2

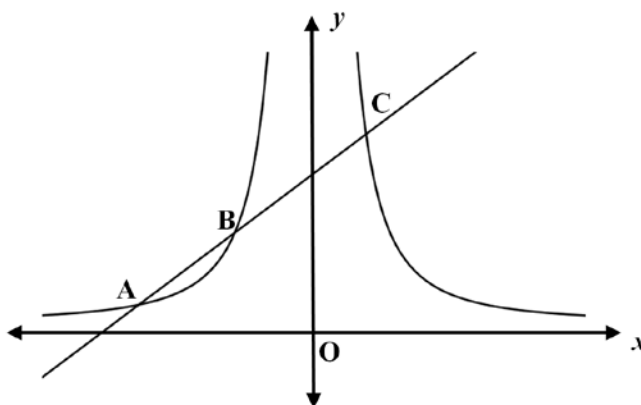
iii) Hence, or otherwise, graph all the roots of $z^6 - 1 = 0$ on an Argand diagram 2

c) Find $\int \frac{dx}{x\sqrt{x^2 - 1}}$, using $x = \sec \theta$ 2

d) Show that $\int_0^1 \frac{\sqrt{x}}{1+x} dx = 2 - \frac{\pi}{2}$ 3

Question 12 (15 marks) Use a SEPARATE writing sheet.

a)



In the diagram above, the points A, B and C represent the points of intersection of the line $y = 4x + 8$ and the curve $y = \frac{1}{x^2}$. The x-values of A, B and C are α , β and γ .

(i) Show that α, β and γ satisfy $4x^3 + 8x^2 - 1 = 0$ 1

(ii) Find a polynomial with roots α^2, β^2 and γ^2 2

(iii) Find $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$ 3

(iv) Prove that $OA^2 + OB^2 + OC^2 = 132$ where O is the origin 3

b) If w is a complex root of the equation $z^3 = 1$

(i) Show that $1 + w + w^2 = 0$ 1

(ii) Find the value of $(1 + 2w + 3w^2)(1 + 2w^2 + 3w)$ 2

c) Evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{5 + 3\cos x} dx$ 3

Question 13 (15 marks) Use a SEPARATE writing sheet.

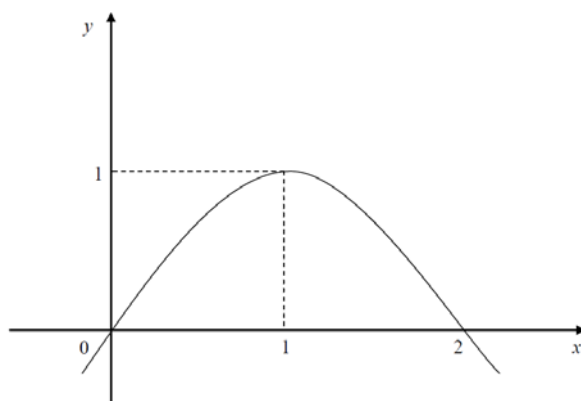
a) By completing the square find $\int \frac{dx}{\sqrt{6x - x^2}}$ 2

b) (i) Find real constants A , B and C such that

$$\frac{x+4}{x(x^2+4)} \equiv \frac{A}{x} + \frac{Bx+C}{x^2+4}$$
 2

(ii) Hence find $\int \frac{x+4}{x(x^2+4)} dx$ 2

c) The graph below shows the curve $y = f(x)$ where $f(x) = x(2-x)$



Without the use of calculus, sketch the following curves. Show any intercepts, asymptotes, end points and turning points. (Use $\frac{1}{3}$ of a page per sketch)

(i) $y = f(2x)$ 1

(ii) $y = \frac{1}{f(x)}$ 2

(iii) $|y| = f(x)$ 2

(iv) $y = \ln f(x)$ 2

(v) $y = f(e^x)$ 2

Question 14 (15 marks) Use a SEPARATE writing sheet.

a) On an Argand diagram let $A = 3 + 4i$

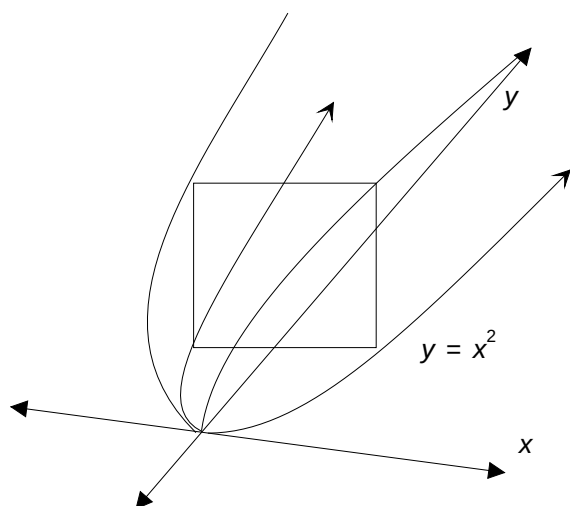
(i) Draw a clear sketch to show the important features of the curve defined by $|z - A| = 5$ 2

(ii) Also for z on this curve, find the maximum value of $|z|$ 1

b) (i) If $I_n = \int_0^{\infty} e^{-x} \sin^n x \, dx$, then prove that $I_n = \frac{n(n-1)}{n^2+1} I_{n-2}$ 3

(ii) Hence evaluate $\int_0^{\infty} e^{-x} \sin^4 x \, dx$ 3

c) Shown below is a solid which has as its base the parabola $y = x^2$ in the XY plane. Sections taken perpendicular to the Y axis are squares. The length of the figure is 9 units.

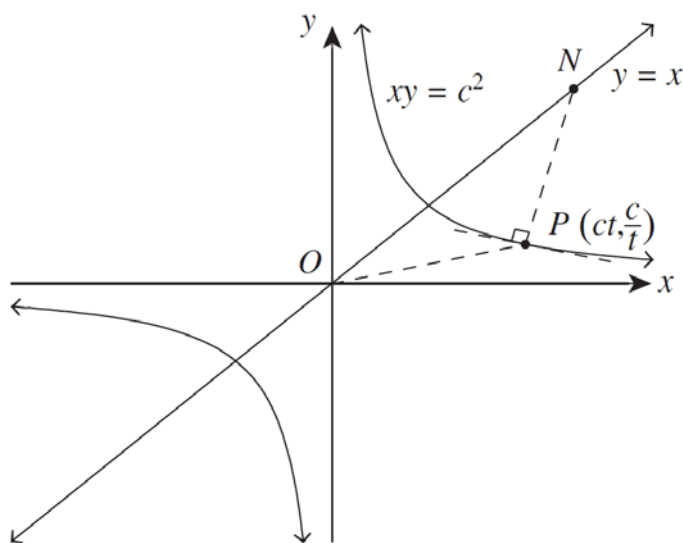


By using the technique of slicing, determine the volume of this solid. 3

d) If $0 < x < y < \frac{1}{2}$ prove that: $\sqrt{xy} < x + y < \sqrt{x+y}$ 3

Question 15 (15 marks) Use a SEPARATE writing sheet.

- a) The diagram below shows the hyperbola $xy = c^2$. The point $P\left(ct, \frac{c}{t}\right)$ lies on the curve where $t \neq 0$. The normal at P intersects the straight line $y = x$ at N , O is the origin.



- | | | |
|-------|---|---|
| (i) | Prove that the equation of the normal at P is $y = t^2x + \frac{c}{t} - ct^3$ | 2 |
| (ii) | Find the coordinates of N | 1 |
| (iii) | Show that triangle OPN is isosceles | 2 |

- b) The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ cuts the x -axis at the points M and N .

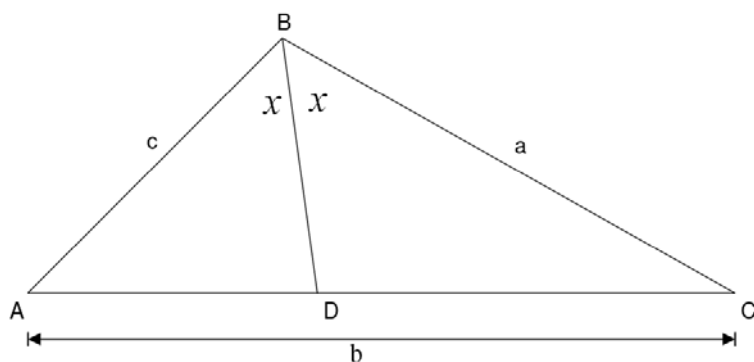
A tangent drawn to the ellipse at a point $P(a \cos t, b \sin t)$ meets the tangents at M and N at the points Q and R respectively.

Given that the equation of the tangent at P is $\frac{x \cos t}{a} + \frac{y \sin t}{b} = 1$ (DO NOT PROVE THIS)

Prove that, for all positions of P , $MQ \times NR$ is a constant 2

- | | | |
|----|---------------------------------------|---|
| c) | Evaluate $\int_0^1 \sin^{-1} x \, dx$ | 3 |
|----|---------------------------------------|---|

d)



In $\triangle ABC$, BD bisects $\angle ABC$ as shown in the diagram.

i) By considering the area of $\triangle ABC$, show that

2

$$BD = \frac{2ac \cos x}{a + c}$$

ii) Show that: $\cos x = \frac{1}{2} \sqrt{\frac{(a + c)^2 - b^2}{ac}}$

2

iii) Hence show that $BD = \frac{\sqrt{ac}}{a + c} \sqrt{(a + c)^2 - b^2}$

1

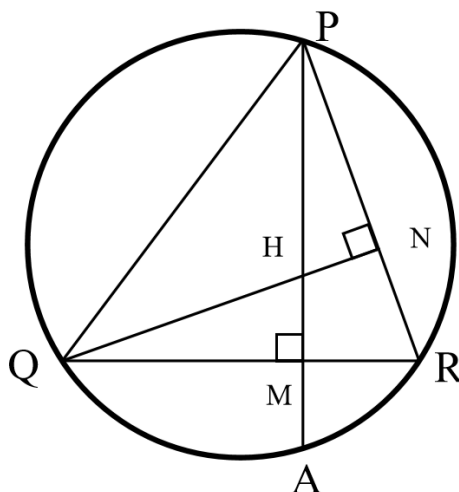
Question 16 (15 marks) Use a SEPARATE writing sheet.

- a) Scientists use a pressure gauge which measures depth as it sinks towards the ocean floor. The gauge of mass 2 kg is released from rest at the ocean's surface. As it sinks in a vertical line, the water exerts a resistance to its motion of $4v$ Newtons, where $v \text{ ms}^{-1}$ is the velocity of the gauge.

Let x be the displacement of the ball measured vertically downwards from the ocean's surface, t be the time in seconds elapsed after the gauge is released, and g be the constant acceleration due to gravity.

- (i) Show that $\frac{d^2x}{dt^2} = g - 2v$ 2
- (ii) Hence show that $t = \frac{1}{2} \log_e \left(\frac{g}{g - 2v} \right)$ 3
- (iii) Show that $v = \frac{g}{2} (1 - e^{-2t})$ 2
- (iv) Write down the limiting (terminal) velocity of the gauge 1
- (v) At a particular location, the gauge takes 180 seconds to hit the ocean floor. 3
Using $g = 10 \text{ ms}^{-2}$, calculate the depth of the ocean at that location, giving your answer correct to the nearest metre.

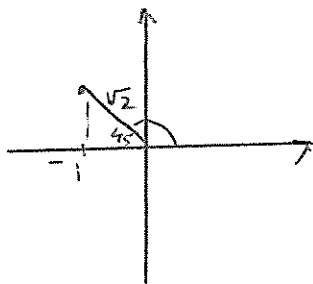
- b) The altitudes PM and QN of an acute angled triangle PQR meet at H. PM produced cuts the circle PQR at A. Copy the diagram onto your answer sheet.



- i) Explain why PQMN is a cyclic quadrilateral 1
- ii) Hence or otherwise prove that $HM = MA$. 3

Multiple Choice

Q1



B

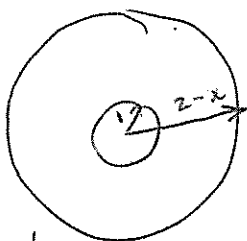
Q2

D

Q3

A

Q4



$$\begin{aligned}
 V &= \int_{-1}^1 \pi [R^2 - r^2] dy \\
 &= \int_{-1}^1 \pi [(2-x)^2 - 1^2] dy \\
 &= \int_{-1}^1 \pi [(2+y^2)^2 - 1^2] dy \\
 &= \int_{-1}^1 \pi [4 + 4y^2 + y^4 - 1] dy \\
 &= \int_{-1}^1 \pi (y^4 + 4y^2 + 3) dy
 \end{aligned}$$

B

Q5 $a^2 = b^2(e^2 - 1)$

$a^2 = 12 \quad b^2 = 8$

$a = 2\sqrt{3} \quad b = 2\sqrt{2}$

$12 = 8(e^2 - 1)$

$\frac{12}{8} = e^2 - 1$

$e^2 = \frac{20}{8}$

$e = \frac{2\sqrt{5}}{2\sqrt{2}} = \frac{\sqrt{5}}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{10}}{2}$

Foci $(0, \pm be)$

$(0, \pm 2\sqrt{2} \times \frac{\sqrt{10}}{2})$

$= (0, \pm \sqrt{20})$

$= (0, \pm 2\sqrt{5})$

C

Q6 $\alpha = 1 + \sqrt{2}i \quad \beta = 1 - \sqrt{2}i \quad \gamma = ?$

$\alpha + \beta = 2$

$\alpha + \beta + \gamma = 0 = -\frac{b}{a} \quad \text{--- (1)}$

$\alpha\beta = 3$

$\alpha\beta\gamma = -\frac{d}{a} = -q \quad \text{--- (2)}$

$\alpha\beta + \beta\gamma + \alpha\gamma = +p \quad \text{--- (3)}$

(1) $2 + \gamma = 0$

$\gamma = -2$

(3) $\alpha\beta\gamma = 3(-2) = -q$

$q = 6$

(2) $3 + -2(1 - \sqrt{2}i) + -2(1 + \sqrt{2}i) = +p$

$3 - 2 + 2\sqrt{2}i - 2 - 2\sqrt{2}i = +p$

$3 - 4 = +p$

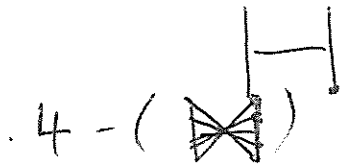
$p = -1$

B

(1)

Q7. All combinations
 ${}^{12}C_3$ -- (12 points choosing 3 vertices)

$3 \cdot {}^5C_3$ -- (3 ~~lines~~ sets of lines containing collinear points)



$4 - ()$
B

Q8 A

Q9 $\int_0^{\frac{\pi}{4}} x \sec^2 x \, dx.$

$$= x \tan x - \int \tan x \, dx$$

$$= [x \tan x + \ln |\cos x|]_0^{\frac{\pi}{4}}$$

$$= \frac{\pi}{4} + \ln\left(\frac{1}{\sqrt{2}}\right)$$

$$= \frac{\pi}{4} - \frac{1}{2} \ln 2.$$

A

Q10. $v \frac{dv}{dx} = \frac{1}{x^3} (6 - 10x)$

$$v \, dv = \frac{1}{x^3} (6 - 10x) \, dx$$

From $v=0, x=1$ to end $v=v, x=x$

$$\int_0^v v \, dv = \int_1^x \left[\frac{1}{x^3} (6 - 10x) \right] dx$$

$$\left[\frac{v^2}{2} \right]_0^v = \left[-\frac{3}{x^2} + \frac{10}{x} \right]_1^x$$

$$\frac{v^2}{2} = -\frac{3}{x^2} + \frac{10}{x} + \frac{3}{1} - 10.$$

$$\frac{v^2}{2} = \frac{1}{x^2} (-3 + 10x - 7x^2)$$

$$v^2 = \frac{1}{x^2} \cdot 2(-3 + 10x - 7x^2)$$

$$v = \pm \frac{1}{x} \sqrt{2(-3 + 10x - 7x^2)}$$

C

Question 11.

$$a) (i) \frac{z}{w} = \frac{3-i}{2+i} \times \frac{2-i}{2-i}$$

$$= \frac{6-3i-2i+i^2}{4-i^2}$$

$$= \frac{5-5i}{5}$$

$$= 1-i$$

$$(ii) \overline{-2iz} = \overline{-2i(3-i)}$$

$$= \overline{-2(3i-i^2)}$$

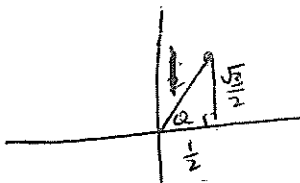
$$= \overline{-2(3i+1)}$$

$$= \overline{-6i-2}$$

$$= -2+6i$$

$$b) z = \frac{1}{2} + \frac{\sqrt{3}}{2}i$$

(i)



$$\text{Mod} = \sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2}$$

$$= 1$$

$$\text{Arg} = \tan^{-1}(\sqrt{3})$$

$$= \frac{\pi}{3}$$

$$z = \text{cis } \frac{\pi}{3}$$

$$(ii) z^6 = (\text{cis } \frac{\pi}{3})^6$$

$$= (\text{cis } 2\pi) \quad \text{by De Moivre's Theorem}$$

$$= 1$$

$$(iii) z^6 - 1 = 0$$

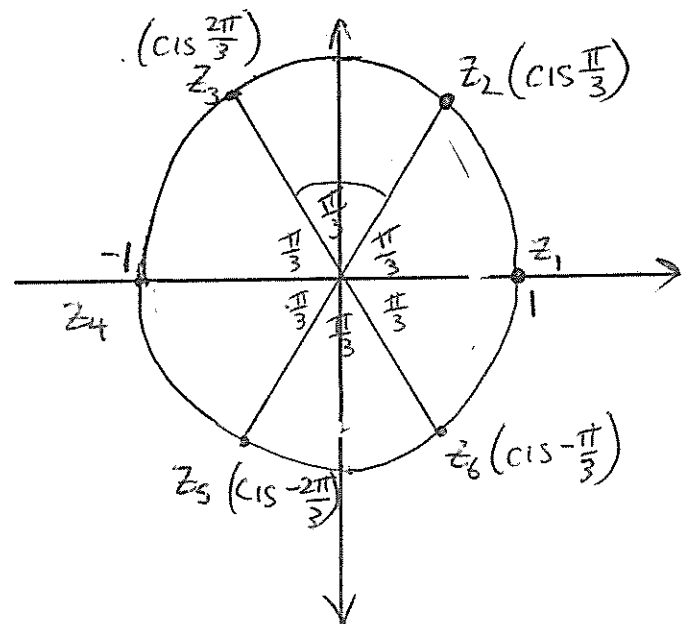
$$z^6 = 1 = \text{cis } (2k\pi), \quad k \text{ integer}$$

$$z = \text{cis } \left(\frac{2k\pi}{6}\right)$$

$$= \text{cis } \left(\frac{k\pi}{3}\right) \quad k=0, \pm 1, \pm 2, -3$$

$$z = 1, -1, \text{cis } \frac{\pi}{3}, \text{cis } \left(-\frac{\pi}{3}\right),$$

$$\text{cis } \left(\frac{2\pi}{3}\right), \text{cis } \left(-\frac{2\pi}{3}\right)$$



$$c) \int \frac{dx}{x\sqrt{x^2-1}} \quad x = \sec \theta$$

$$dx = \sec \theta \tan \theta d\theta.$$

$$\int \frac{\sec \theta \cdot \tan \theta d\theta}{\sec \theta \cdot \tan \theta} = \int d\theta$$

$$= \theta + C$$

$$= \sec^{-1} x + C$$

$$d) \int_0^1 \frac{\sqrt{x}}{1+x} dx = 2 - \frac{\pi}{2}$$

$$\text{Let } u = x^{\frac{1}{2}}$$

$$x=0 \quad u=0$$

$$x=1 \quad u=1.$$

$$\frac{du}{dx} = \frac{1}{2} x^{-\frac{1}{2}}$$

$$du = \frac{1}{2u} dx.$$

$$= \int_0^1 \frac{u \cdot 2u du}{1+u^2}$$

$$= 2 \int \left(\frac{1+u^2-1}{1+u^2} \right) du$$

$$= 2 \int_0^1 du - 2 \int_0^1 \frac{1}{1+u^2} du$$

$$= 2[u]_0^1 - 2[\tan^{-1} u]_0^1$$

$$= 2 - 2 \times \frac{\pi}{4}$$

$$= 2 - \frac{\pi}{2}$$

Question 12

a (i) $y = \frac{1}{x^2}$

$y = 4x + 8$

for intersection points A, B, C

$$4x + 8 = \frac{1}{x^2}$$

$$4x^3 + 8x^2 - 1 = 0$$

(ii) $y = x^2$
 $x = y^{\frac{1}{2}}$

$$4(y^{\frac{1}{2}})^3 + 8(y^{\frac{1}{2}})^2 - 1 = 0$$

$$4y^{\frac{3}{2}} + 8y - 1 = 0$$

$$4y^{\frac{3}{2}} = 1 - 8y$$

$$16y^3 = 1 + 64y^2 - 16y$$

$$16y^3 - 64y^2 + 16y - 1 = 0$$

since y is a dummy variable

$$16x^3 - 64x^2 + 16x - 1 = 0$$

(iii) $\frac{1}{x^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2}{\alpha^2\beta^2\gamma^2}$

$$= \frac{\sum 2 \text{ at a time}}{\text{product of roots}}$$

$$= \frac{1}{\frac{1}{16}}$$

$$= 16.$$

(iv) $OA^2 = \alpha^2 + \left(\frac{1}{\alpha^2}\right)^2 = \alpha^2 + \frac{1}{\alpha^4}$

$$OB^2 = \beta^2 + \frac{1}{\beta^4}$$

$$OC^2 = \gamma^2 + \frac{1}{\gamma^4}$$

$$\therefore OA^2 + OB^2 + OC^2 = \alpha^2 + \beta^2 + \gamma^2 + \frac{1}{\alpha^4} + \frac{1}{\beta^4} + \frac{1}{\gamma^4}$$

Now $\frac{1}{\alpha^4} + \frac{1}{\beta^4} + \frac{1}{\gamma^4} = \frac{\alpha^4\beta^4\alpha^4\gamma^4 + \beta^4\gamma^4}{\alpha^4\beta^4\gamma^4}$

and $\alpha^4\beta^4 + \alpha^4\gamma^4 + \beta^4\gamma^4$

$$= (\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2)^2 - 2\alpha^2\beta^2\gamma^2(\alpha^2\beta^2 + \gamma^2)$$

$$= 1^2 - 2 \times \frac{1}{16} \times 4$$

$$= \frac{1}{2}$$

$$\alpha^4\beta^4\gamma^4 = \frac{1}{256}$$

$$\therefore \alpha^2 + \beta^2 + \gamma^2 + \frac{1}{\alpha^4} + \frac{1}{\beta^4} + \frac{1}{\gamma^4}$$

$$= 4 + \frac{256}{2}$$

$$= 132.$$

b) (i) $z^3 = 1$ w complex root.

$$z^3 - 1 = 0.$$

$$(z-1)(z^2+z+1)=0.$$

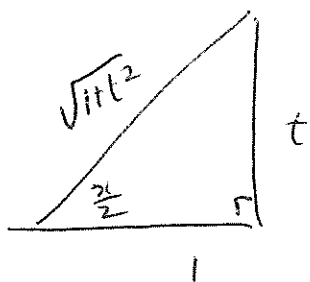
as $w \neq 0$ w root.

$$\therefore w^2 + w + 1 = 0.$$

$$\begin{aligned} \text{(ii)} & (1+2w+3w^2)(1+2w^2+3w) \\ &= [1+2w+3(-1-w)][1+2(-1-w)+3w] \\ &= (1+2w-3-3w)(1-2-2w+3w) \\ &= (-2-w)(-1+w) \\ &= 2-2w+w-w^2 \\ &= 2-w-w^2 \\ &= 2+1 \quad (1+w+w^2=0) \\ &= 3. \end{aligned}$$

$$c) \int_0^{\frac{\pi}{2}} \frac{1}{5+3\cos x} dx$$

$$\text{Let } t = \tan \frac{x}{2}$$



$$\frac{dt}{dx} = \sec^2 \frac{x}{2} \cdot \frac{1}{2}$$

$$\frac{dt}{dx} = \frac{1}{2}(1+t^2)$$

$$\frac{2dt}{1+t^2} = dx \quad \begin{matrix} x=0, t=0 \\ x=\frac{\pi}{2}, t=1 \end{matrix}$$

$$I = \int_0^1 \frac{2 dt}{(1+t^2)(5+3 \times \frac{1-t^2}{1+t^2})}$$

$$= \int_0^1 \frac{2 dt}{5+5t^2+3-3t^2}$$

$$= \int_0^1 \frac{2 dt}{8+2t^2}$$

$$= \int_0^1 \frac{dt}{4+t^2}$$

$$= \frac{1}{2} \left[\tan^{-1} \frac{t}{2} \right]_0^1$$

$$= \frac{1}{2} \tan^{-1} \left(\frac{1}{2} \right)$$

Question 13

a) $\int \frac{dx}{\sqrt{9-9+6x-x^2}}$

$= \int \frac{dx}{\sqrt{9-(x-3)^2}}$

$= \sin^{-1}\left(\frac{x-3}{3}\right) + C$

b(i) $\frac{x+4}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{x^2+4}$

$x+4 = A(x^2+4) + (Bx+C)x$

$x+4 = (A+B)x^2 + Cx + 4A$

$4A=4 \Rightarrow \boxed{A=1}$

$Cx=x \Rightarrow \boxed{C=1}$

$A+B=0 \Rightarrow \boxed{B=-1}$

(ii) $\int \frac{x+4}{x(x^2+4)} dx$

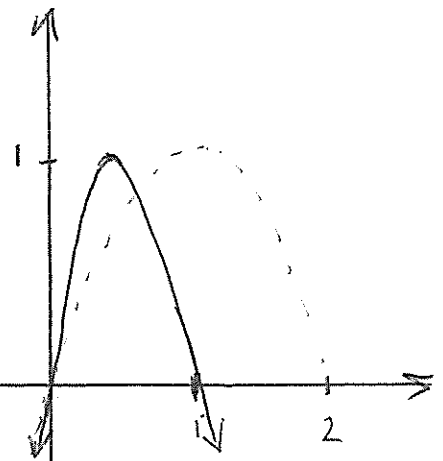
$= \int \frac{1}{x} dx + \int \frac{-x+1}{x^2+4} dx$

$= \int \frac{1}{x} dx - \frac{1}{2} \int \frac{2x}{x^2+4} dx + \int \frac{dx}{x^2+4}$

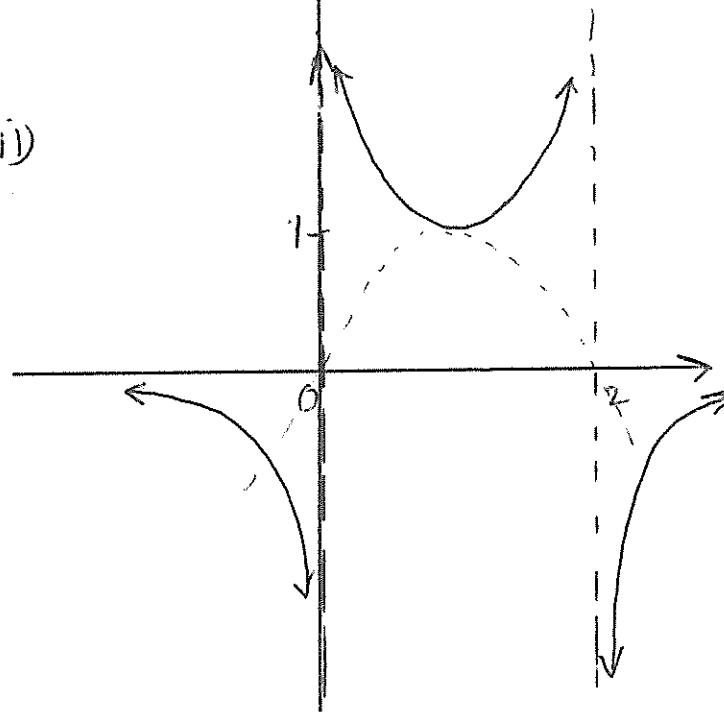
$= \ln(x) - \frac{1}{2} \ln(x^2+4) + \frac{1}{2} \tan^{-1} \frac{x}{2} + C$

$= \ln\left(\frac{x}{\sqrt{x^2+4}}\right) + \frac{1}{2} \tan^{-1} \frac{x}{2} + C$

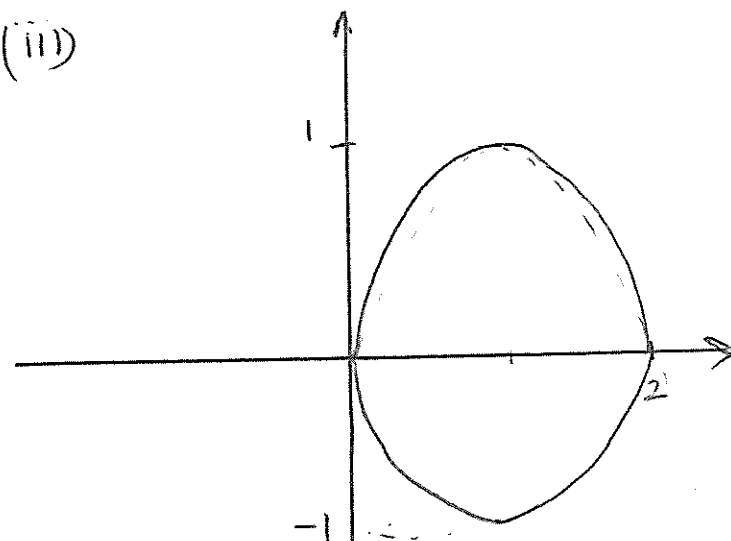
C(i)



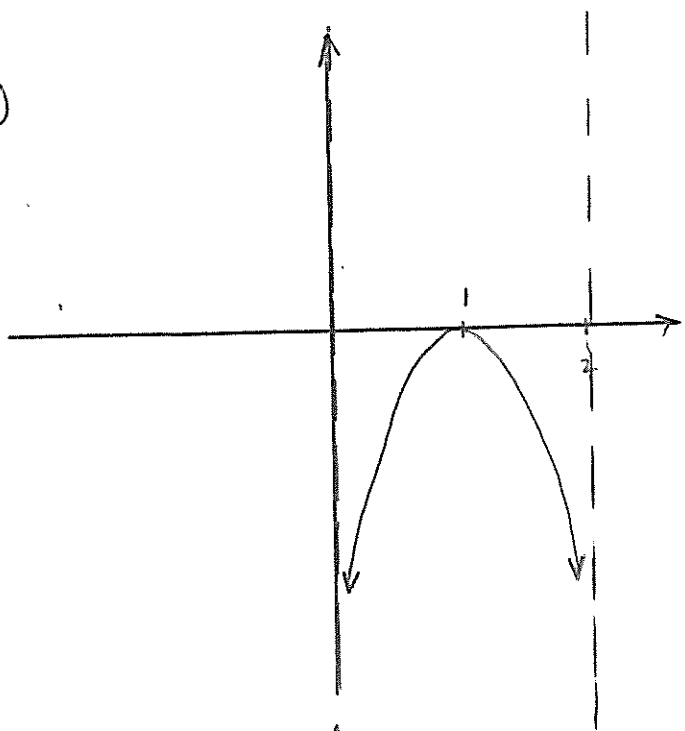
(ii)



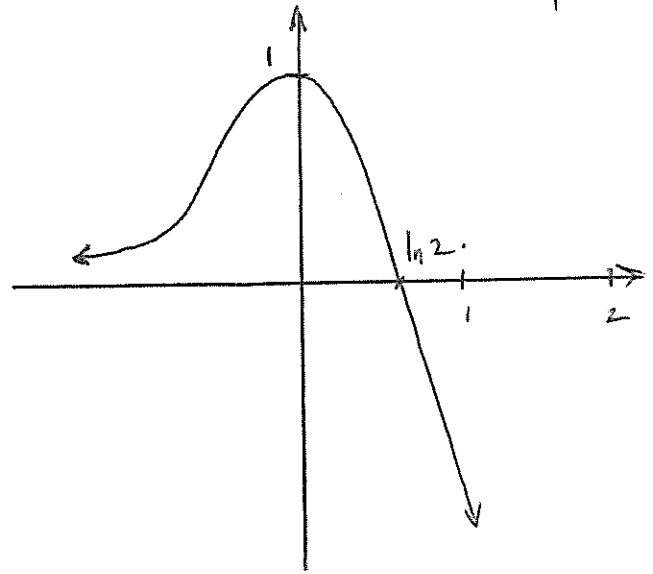
(iii)



(iv)

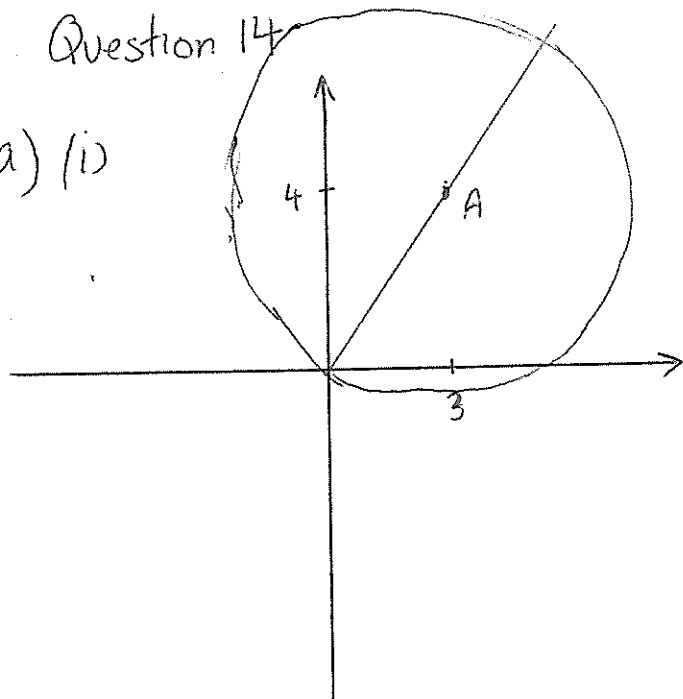


(v)



Question 14

a) (i)



(ii) max |z| line from origin to other side of the circle
 max |z| = 10 as |OA| = 5 units

$$b) (i) \cdot I_n = \int_0^{\infty} e^{-x} \sin^n x \, dx$$

$$= \left[\sin^n x \frac{e^{-x}}{-1} \right]_0^{\infty} - \int_0^{\infty} n \sin^{n-1} x \cdot \frac{\cos x \cdot e^{-x}}{-1} \, dx$$

(integration by parts)
 $(e^{-\infty} = 0)$

$$= n \int_0^{\infty} (\sin^{n-1} x \cos x) e^{-x} \, dx$$

$$= n \left[\left(\sin^n x \cos x \frac{e^{-x}}{-1} \right) - \int_0^{\infty} [(n-1) \sin^{n-2} x \cos^2 x - \sin^{n-2} x \sin x] \frac{e^{-x}}{-1} \, dx \right]$$

$$= n(n-1) \int_0^{\infty} \sin^{n-2} x (1 - \sin^2 x) e^{-x} \, dx - n \int_0^{\infty} \sin^{n-2} x \cdot e^{-x} \, dx$$

$$I_n = n(n-1) I_{n-2} - n(n-1) I_n - n I_n$$

$$I_n + (n^2 - n + n) I_n = n(n-1) I_{n-2}$$

$$(n^2 + 1) I_n = n(n-1) I_{n-2}$$

$$I_n = \frac{n(n-1)}{n^2 + 1} I_{n-2}$$

$$(ii) \int_0^{\infty} e^{-x} \sin^4 x \, dx = I_4$$

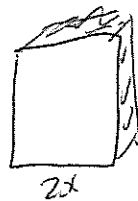
$$I_4 = \frac{4(4-1)}{4^2 + 1} I_2$$

$$I_2 = \frac{2(2-1)}{2^2 + 1} I_0$$

$$= \frac{2}{5} \times 1 = \frac{2}{5}$$

$$I_4 = \frac{4 \times 3}{17} \times \frac{2}{5} = \frac{24}{85}$$

$$c) \int V = \sum_{\delta y} (2x)^2 \delta y$$



$$V = \int_0^9 4x^2 \, dy$$

$$= \int_0^9 4y \, dy$$

$$= \left[\frac{4y^2}{2} \right]_0^9$$

$$= 162 \, u^3$$

$$d) (\sqrt{x} - \sqrt{y})^2 \geq 0$$

$$x + y - 2\sqrt{xy} \geq 0$$

$$x + y \geq 2\sqrt{xy}$$

$$\therefore x + y \geq \sqrt{xy} \dots \dots (1)$$

If $x \leq \frac{1}{2}$ and $y \leq \frac{1}{2}$ then
 $x + y \leq 1$

$$\therefore \sqrt{xy} \geq x + y \dots \dots (2)$$

as the square root of a number between 0 and 1 is greater than the number.

$\therefore$ from (1) and (2)

$$\sqrt{xy} \leq x + y \leq \sqrt{xy}$$

Question 15

$$\begin{aligned} \text{a) } xy &= c^2 \\ xy' + y &= 0 \\ y' &= -\frac{y}{x} \end{aligned}$$

$$= \frac{-c}{t \times t} = -\frac{1}{t^2}$$

$\therefore$ m of the normal at $P = t^2$

equation of the normal

$$y - \frac{c}{t} = t^2(x - ct)$$

$$y = t^2x + \frac{c}{t} - ct^2$$

(ii) for N, $y = x$

$$x = t^2x + \frac{c}{t} - ct^2$$

$$x(1 - t^2) = \frac{c}{t}(1 - t^2)$$

$$x = \frac{c(1+t^2)(1-t^2)}{t(1-t^2)}$$

$$x = \frac{c(1+t^2)}{t}$$

$$y = \frac{c(1+t^2)}{t}$$

$$\begin{aligned} \text{(iii) } |OP| &= \sqrt{(ct)^2 + \left(\frac{c}{t}\right)^2} \\ &= \frac{c}{t} \sqrt{t^4 + 1} \end{aligned}$$

$$|PN| = \sqrt{\left(ct - \frac{c(1+t^2)}{t}\right)^2 + \left(\frac{c}{t} - \frac{c(1+t^2)}{t}\right)^2}$$

$$= \frac{c}{t} \sqrt{(t^2 - 1 - t^2)^2 + (1 - 1 - t^2)^2}$$

$$= \frac{c}{t} \sqrt{1 + t^4} \quad \therefore |OP| = |PN|$$

$\triangle OPN$ is an isosceles triangle

b) since the tangent at M and N are vertical lines
 $\therefore$ x-coordinates of Q and R are a, -a.

$$Q_y = (1 - \cos t) \frac{b}{\sin t}$$

$$R_y = (1 + \cos t) \frac{b}{\sin t}$$

$$MQ \times NR = (1 - \cos t) \frac{b}{\sin t} \times (1 + \cos t) \frac{b}{\sin t}$$

$$= (1 - \cos^2 t) \frac{b^2}{\sin^2 t}$$

$$= b^2 \text{ (which is a constant)}$$

$$\text{c) } \int_0^1 \sin^{-1} x \, dx$$

$$= \left[\sin^{-1} x \cdot x \right]_0^1 - \int_0^1 \frac{x}{\sqrt{1-x^2}} \, dx$$

$$= \frac{\pi}{2} - \frac{1}{2} \int_0^1 \frac{2x}{\sqrt{1-x^2}} \, dx$$

$$u = 1 - x^2$$

$$\frac{du}{dx} = -2x$$

$$du = -2x \, dx$$

$$= \frac{\pi}{2} + \frac{1}{2} \left[\frac{u^{-\frac{1}{2}+1}}{\frac{1}{2}} \right]$$

$$= \frac{\pi}{2} + \left[\sqrt{1-x^2} \right]_0^1$$

$$= \frac{\pi}{2} + 0 - 1$$

$$= \frac{\pi}{2} - 1$$

$$(1) \text{ Area of } \triangle ABC = \frac{1}{2}ac \sin 2x$$

$$= \frac{1}{2}ac \times 2 \sin x \cos x$$

$$= ac \sin x \cos x$$

$$\text{Area of } \triangle ABD = \frac{1}{2}c \times BD \sin x$$

$$\text{Area of } \triangle BDC = \frac{1}{2}BD \times a \sin x$$

$$\therefore ac \sin x \cos x = \frac{1}{2}BD(c \sin x + a \sin x)$$

$$\frac{2ac \cos x}{a+c} = BD$$

$$(ii) \cos 2x = \frac{a^2+c^2-b^2}{2ac}$$

$$2\cos^2 x - 1 = \frac{a^2+c^2-b^2}{2ac}$$

$$2\cos^2 x = \frac{a^2+c^2-b^2}{2ac} + 1$$

$$= \frac{(a+c)^2 - b^2}{2ac}$$

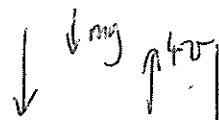
$$\cos^2 x = \frac{1}{4} \left[(a+c)^2 - b^2 \right]$$

$$\cos x = \frac{1}{2} \sqrt{\frac{(a+c)^2 - b^2}{ac}}$$

$$\text{Hence } BD = \frac{2ac}{a+c} \times \frac{1}{2} \sqrt{\frac{(a+c)^2 - b^2}{ac}}$$

$$BD = \frac{\sqrt{ac}}{a+c} \sqrt{(a+c)^2 - b^2}$$

Question 16



$$a(i) \quad m\ddot{x} = mg - 4v, \quad m = 2 \text{ kg}$$

$$\ddot{x} = g - 2v$$

$$(ii) \quad \ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{dv}{dt} = v \frac{dv}{dx}$$

$$\frac{dv}{dt} = g - 2v$$

$$\int_0^v \frac{dv}{g-2v} = \int_0^t dt$$

$$-\frac{1}{2} [\ln(g-2v)]_0^v = t$$

$$\Rightarrow t = -\frac{1}{2} (\ln(g-2v) - \ln g)$$

$$t = \frac{1}{2} \ln \left(\frac{g}{g-2v} \right)$$

$$(iii) \quad 2t = \ln \frac{g}{g-2v} \quad (\text{from ii})$$

$$e^{2t} = \frac{g}{g-2v}$$

$$g-2v = ge^{-2t}$$

$$2v = g - ge^{-2t}$$

$$v = \frac{g}{2} (1 - e^{-2t})$$

$$(iv) \quad \ddot{x} \rightarrow 0 \Rightarrow g - 2v \rightarrow 0$$

$$\Rightarrow v \rightarrow \frac{g}{2}$$

OR.

$$t \rightarrow \infty, e^{-2t} \rightarrow 0 \Rightarrow v \rightarrow \frac{g}{2}$$

(part iii)

$$(v) \quad v = \frac{g}{2} (1 - e^{-2t})$$

$$g = 10$$

$$v = 5(1 - e^{-2t})$$

$$\int dx = \int 5(1 - e^{-2t}) dt$$

$$x = 5 \left[t - \frac{e^{-2t}}{-2} \right] + C$$

$$= 5 \left[t + \frac{e^{-2t}}{2} \right] + C$$

$$x=0 \quad t=0$$

$$0 = 5 \left(\frac{1}{2} \right) + C$$

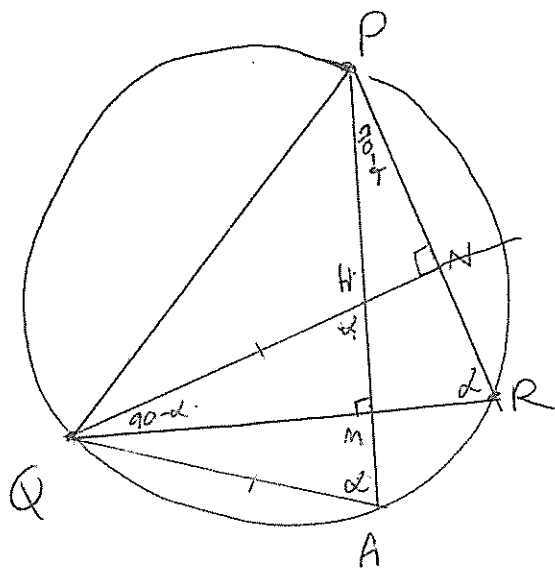
$$C = -\frac{5}{2}$$

$$x = 5 \left(t + \frac{e^{-2t}}{2} \right) - \frac{5}{2}$$

$$t = 180 \quad x = ?$$

$$x = 5 \left(180 + \frac{e^{-360}}{2} \right) - \frac{5}{2}$$
$$\approx 898 \text{ m (nearest metre)}$$

b(i)



PQMN is a cyclic quadrilateral
 PQ subtends 90° at M and N
 $\therefore$ PQMN is a cyclic
 quadrilateral with PQ
 as diameter

(ii) $AM = MA$

Let $\angle PRQ = \alpha$

$\therefore \angle PRQ = \angle PAQ = \alpha$

($\angle$'s on circumference subtended
 by arc PQ, circle PQR)

$\angle RPA = 90 - \alpha$ ($\angle$ sum of $\triangle PRM$)

$\angle RPA = \angle NQR = 90 - \alpha$

($\angle$'s on circumference subtended
 by arc MN, cyclic PQMN)

$\therefore \angle QHM = \alpha$ ($\angle$ sum of $\triangle QHM$)

$\therefore \angle QHM = \angle QAH = \alpha$

$\therefore HQ = QA$ ($\angle$'s opposite equal
 sides $\triangle QHM$)

$\therefore HM = MA$

($\triangle QMH \cong \triangle QMA$)

QM common

$\angle QHA = \angle QAN = \alpha$

$QH = QA$
 (shown)